Waverider Design Tool

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May 2, 2026

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1 Introduction

This tool implements an inverse method for generating waverider geometries at a prescribed design Mach number, given a trailing edge profile.

2 Flight trajectory and mission analysis

3 Cone-derived Waverider Geometry Generation

In this section, we describe our approach to generating a waverider geometry from a shock-generating cone. It consists of three major steps: (1) solving the inviscid flowfield around the cone via the Taylor-Maccoll equations, (2) defining the trailing edge curve and projecting it onto the shock cone to obtain the leading edge, and (3) tracing streamlines from the leading edge to the trailing edge to carve out the upper and lower surfaces. Note that the algorithm proposed in this section largely follows that described by [5], and builds on the partial python implementation provided by [11].

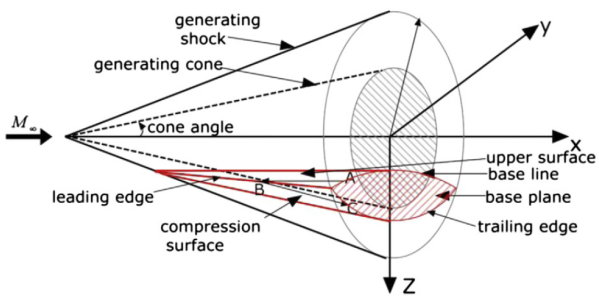


Figure 1: Schematic of the cone-derived waverider geometry generation algorithm. The shock-generating cone is shown in black dashed lines, with the shock surface in solid lines. The trailing edge (TE) is defined in the base plane $x = L$ and parameterized by $R_{1,\text{frac}}$, W_{frac}, and n. The leading edge (LE) is obtained by projecting the TE points radially inward along the shock cone surface. The lower surface is generated by tracing streamlines from the LE to the TE, while the upper surface is a ruled flat top. Adapted from [11].

3.1 Inviscid flow around a shock-generating cone

Cone-derived waveriders are generated on the axisymmetric conical shocksurface (with half-angle β) produced by a sharp-nosed cone (with half-angle θ_c and length L). This choice is convenient as the shock geometry and inviscid flowfield behind the shock are well-characterized via the oblique shock relations and the Taylor-Maccoll equations [6], respectively. These latter are a set of ODEs governing the flow in the conical region between the shock and the body cone, which exploit the self-similar (conical) nature of the flowfield — all flow properties depend only on the polar angle θ measured from the cone axis. Ahead of the shock, the flowfield is simply the free-stream flow.

3.1.1 Shock-jump relations

For an oblique shock, relationship between β, M_∞, and θ_c is given by the $\theta - \beta - M$ relation:

$$\tan\theta_c = 2\cot\beta\,\frac{M_\infty^2\sin^2\beta - 1}{M_\infty^2(\gamma + \cos 2\beta) + 2}. \tag{1}$$

The normal component of the upstream Mach number is $M_{n1} = M_\infty \sin\beta$ and the post-shock normal Mach number is given by the Rankine-Hugoniot jump conditions:

$$M_{n2}^2 = \left(M_{n1}^2 + \frac{2}{\gamma-1}\right)\left(\frac{2\gamma}{\gamma-1}M_{n1}^2 - 1\right)^{-1}. \tag{2}$$

The total post-shock Mach number follows from the geometry of the shock:

$$M_2 = \frac{M_{n2}}{\sin(\beta-\theta_c)}. \tag{3}$$

3.1.2 Taylor-Maccoll conical flow

For calorically-perfect, steady, inviscid, isentropic flow about an axisymmetric cone with a perfectly self-similar (conical) structure, the velocity field in spherical coordinates (r, θ, ϕ) reduces to $\mathbf{V} = V_r(\theta)\,\hat{r} + V_\theta(\theta)\,\hat{\theta}$. In the Taylor-Maccoll formulation, the velocity is non-dimensionalized by the maximum attainable speed, *i.e.,* $V' = V/V_{\max}$, which corresponds to complete conversion of enthalpy to kinetic energy in an adiabatic process:

$$V_{\max} = \sqrt{\frac{2c_pT_0}{\gamma-1}\cdot(\gamma-1)} = \sqrt{\frac{2\gamma RT_0}{\gamma-1}}, \tag{4}$$

where $T_0 = T_\infty\big(1 + \frac{\gamma-1}{2}M_\infty^2\big)$, and is in turn related to the local Mach number through the energy equation:

$$\frac{1}{2}V'^2 + \frac{1}{\gamma-1}\frac{a^2}{V_{\max}^2} = \frac{1}{2} \quad\Longrightarrow\quad V' = \left(\sqrt{\frac{2}{(\gamma-1)M^2}+1}\right)^{-1/2} \tag{5}$$

where a is the local speed of sound.

Based on the axisymmetric, steady, irrotational (and constant entropy, via Crocco's theorem) nature of the flow behind the shock, the Euler equations reduce to the following non-dimensional governing equations [6]:

$$\frac{\gamma-1}{2}\left(1 - V_r'^2 - V_\theta'^2\right)\left(2V_r' + \frac{V_\theta'}{\tan\theta}\right) - V_r'V_\theta'^2 = V_\theta'\left(V_r'V_\theta' - \frac{\gamma-1}{2}(1 - V_r'^2 - V_\theta'^2)\cot\theta\right)' \tag{6}$$

Introducing $A = V_r'$ and $A' = \mathrm{d}V_r'/\mathrm{d}\theta$ (with the conical kinematics giving $V_\theta' = A'$), the equation is rewritten as an explicit second-order ODE for A:

$$A'' = \frac{A\,A'^2 - \frac{\gamma-1}{2}(1 - A^2 - A'^2)\left(2A + \frac{A'}{\tan\theta}\right)}{\frac{\gamma-1}{2}(1 - A^2 - A'^2) - A'^2} \tag{7}$$

This is split into the first-order system:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\begin{pmatrix} A \\ A' \end{pmatrix} = \begin{pmatrix} A' \\ A'' \end{pmatrix} \tag{8}$$

We solve this system by integrating inward from the shock surface $\theta = \beta$, where initial conditions are set by the oblique-shock solution:

$$V_{r,i}' = V'\cos(\beta-\theta_c) \tag{9}$$

$$V_{\theta,i}' = -V'\sin(\beta-\theta_c), \tag{10}$$

toward the cone axis ($\theta \to 0$). The body cone, located at θ_c, requires the streamlines to lie on the cone surface, *i.e.*, $V'_\theta = A' = 0$. The ODE is solved with `scipy.integrate.solve_ivp` using the Runge-Kutta 4,5 (RK45) method. This fully determines the flowfield around the generating cone.

3.2 Determination of trailing and leading edge profiles

The **trailing edge (TE)** lies in the base plane $x = L$. We define it via three parameters: $R_{1,\text{frac}}$, W_{frac}, and n, such that any valid combination of these parameters results in a physically meaningful trailing-edge curve: one that remains inside the shock circle of radius $R_s = L \tan\beta$, is smooth, and remains below the cone axis. The parameters are defined as

$$R_1 = R_{1,\text{frac}}\, R_s, \qquad y_{\text{up}} = W_{\text{frac}}\, R_s, \tag{11}$$

with bounds $0.20 \le R_{1,\text{frac}} \le 0.85$ and $0.25 \le W_{\text{frac}} \le 0.85$. We define a gap function in squared-radius form,

$$g(y) = (R_s^2 - R_1^2)\left(1 - \frac{y^2}{y_{\text{up}}^2}\right)^n \quad \text{for} \quad |y| \le y_{\text{up}}, \tag{12}$$

where $n = n_{\text{shape}}$ (bounded to $0.6 \le n \le 10$) controls the roundness. The TE curve is then

$$z_{\text{TE}}(y) = -\sqrt{R_s^2 - y^2 - g(y)}, \tag{13}$$

which enforces $z_{\text{TE}}(0) = -R_1$ and $z_{\text{TE}}(y_{\text{up}}) = -\sqrt{R_s^2 - y_{\text{up}}^2}$ so the curve meets the shock circle at the tip. Larger n_{shape} makes the tip tangent to the shock; $n_{\text{shape}} < 1$ produces a sharper tip. An example of the trailing edge curve is shown in Fig. 2.

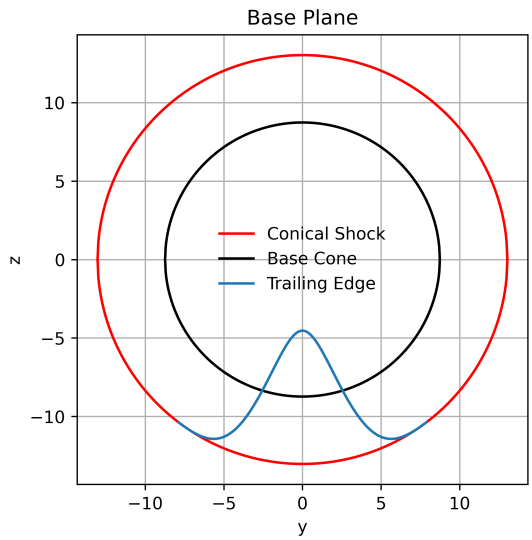


Figure 2: Example trailing edge curve for $M_\infty = 6, \beta = 13.791°, R_{1,\text{frac}} = 0.348, W_{\text{frac}} = 0.605$, and $n = 4.215$. The shock circle is shown for context.

The **leading edge (LE)** is obtained by projecting every point on the TE curve radially inward along the shock cone surface. For a point $(y, z) = (y_{\text{TE}}, z_{\text{TE}})$ in the base plane, the projection maintains the same azimuthal direction while scaling the axial coordinate to keep the point on the cone. The radial distance from the cone axis is $\rho = \sqrt{y^2 + z^2}$; because the shock cone satisfies $\rho = x \tan\beta$, the axial coordinate of the LE is:

$$X_{\text{LE}} = L\left(1 - \frac{R_s - \rho}{R_s}\right) = \frac{L\,\rho}{R_s} \tag{14}$$

with $Y_{\text{LE}} = y_{\text{TE}}$ and $Z_{\text{LE}} = z_{\text{TE}}$.

3.3 Generation of the waverider surfaces via streamline tracing

The full 3D waverider surface is generated by tracing streamlines in the conical flowfield. The upper surface is designed as a ruled surface swept from leading edge straight to the trailing edge at constant (y, z) — a flat top that sees only the freestream. Note that an improvement could be made by instead allowing for a mild expansion along the upper surface which would reduce pressure and allow for slightly more lift; this has been considered in the literature, but as it results in only a few percent improvement in L/D [2], we neglect it as a second-order consideration. To compute the lower surface, streamlines are seeded just under the leading edge, and each is traced inward until it intersects the base plane. A streamline in the conical flowfield satisfies:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = r\,\frac{V_r'(\theta)}{V_\theta'(\theta)} \tag{15}$$

where $V_r'(\theta)$ and $V_\theta'(\theta)$ are provided by the Taylor-Maccoll solution. Equation (15) is solved via forward Euler.

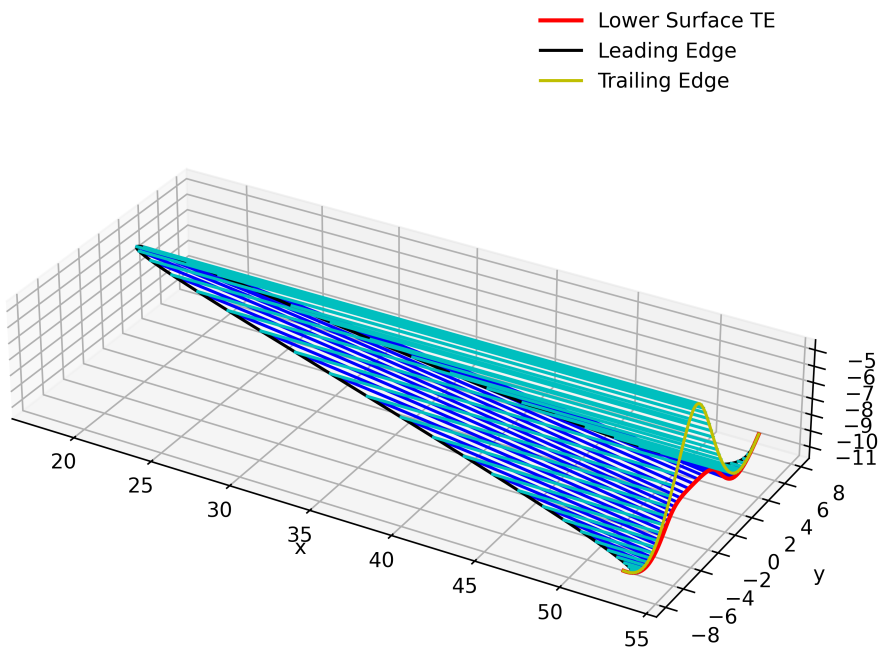


Figure 3: Example streamline tracing from the leading edge to the trailing edge, for $M_\infty = 6, \beta = 13.791°, R_{1,\mathrm{frac}} = 0.348, W_{\mathrm{frac}} = 0.605$, and $n = 4.215$.

Starting from (r_0, θ_0) at the shock surface, successive points are computed as:

$$r_{i+1} = r_i + \frac{V_r'(\theta_i)}{V_\theta'(\theta_i)}\, r_i\, \Delta\theta \tag{16}$$

where $\Delta\theta = \theta_{i+1} - \theta_i < 0$. Integration stops when the streamline crosses the base plane $x = L$, *i.e.,* when $r_i \cos\theta_i < L \leq r_{i+1}\cos\theta_{i+1}$. At that crossing a linear interpolation gives the precise intersection point. The interpolated z-coordinate is:

$$z_k^* = \cos\phi_k \left[\frac{r_{i+1}\sin\theta_{i+1} - r_i \sin\theta_i}{r_{i+1}\cos\theta_{i+1} - r_i\cos\theta_i}\, L + r_i \sin\theta_i - r_i\cos\theta_i\, \frac{r_{i+1}\sin\theta_{i+1} - r_i\sin\theta_i}{r_{i+1}\cos\theta_{i+1} - r_i\cos\theta_i} \right], \tag{17}$$

where the azimuthal angle of the plane is $\phi_k = -\arctan\left(\frac{Y_k}{Z_k}\right)$. From z_k^* the intersection spherical

coordinates are recovered:

$$r_k^* = \sqrt{(z_k^*/\cos\phi_k)^2 + L^2}, \qquad \theta_k^* = \arctan\left(\frac{z_k^*/\cos\phi_k}{L}\right). \tag{18}$$

Fig. 3 shows an example of the streamlines traced from the leading edge to the trailing edge for the same case as Fig. 2, which form the lower surface of the waverider. The upper surface is generated by connecting each point on the leading edge to its corresponding point on the trailing edge via a straight line, forming a ruled surface.

3.4 Algorithm summary

Step 1. Input: Read freestream conditions (M_∞, γ), generating shock parameters (β, L), and the trailing edge description $(R_{1,\text{frac}}, W_{\text{frac}}, n)$.

Step 2. Oblique shock: Solve Eqs. (1) and (3) to obtain M_2, θ; form initial conditions $(V'_{r,i}, V'_{\theta,i})$ via Eqs. (9) and (10).

Step 3. Taylor-Maccoll: Integrate Eq. (7) from $\theta = \beta$ inward with RK45 until $V'_\theta = 0$; record body-cone angle θ_c.

Step 4. Trailing edge: Evaluate the admissible back-face curve Eq. (13) over $|y| \leq y_{\text{up}}$; compute break-point array.

Step 5. Leading edge: Project TE points onto shock cone via Eq. (14).

Step 6. Streamline tracing: For each of N_l seed points on the leading edge:

a. compute plane azimuth ϕ,

b. re-integrate Taylor-Maccoll on uniform θ-grid,

c. Euler-march Eq. (16) until base-plane crossing,

d. interpolate exact intersection Eq. (17),

e. convert to Cartesian coordinates.

Step 7. B-spline: Fit smooth curve through base-plane endpoints.

Step 8. Upper surface: Ruled lines from LE break points to TE.

Example waverider geometries generated by this algorithm for fixed M_∞, L, β are shown in Fig. 4, showing that diverse shapes can be generated by varying the input parameters.

4 Aerothermodynamic Characterization of a given Waverider

4.1 Pressure distribution and inviscid forces

By design, the **upper surface** of the waverider is a ruled flat top that sees only the undisturbed freestream. Though the blunt edge of a real waverider would produce a weak detached bow shock and a more complex flowfield, we neglect this consideration in the present analysis. Thus, we take the pressure coefficient over the upper surface is zero ($C_p^{\text{upper}} = 0$), and assume that it contributes no pressure drag (but that it will contribute to the skin friction drag, which is calculated in the next section).

Conversely, the pressure coefficient on the **lower surface** is non-zero, and is computed from the Taylor-Maccoll solution. Because the lower surface is a stream surface, each streamline lies in an azimuthal plane and its boundary-layer-edge properties are functions of the polar angle θ from the cone axis. For

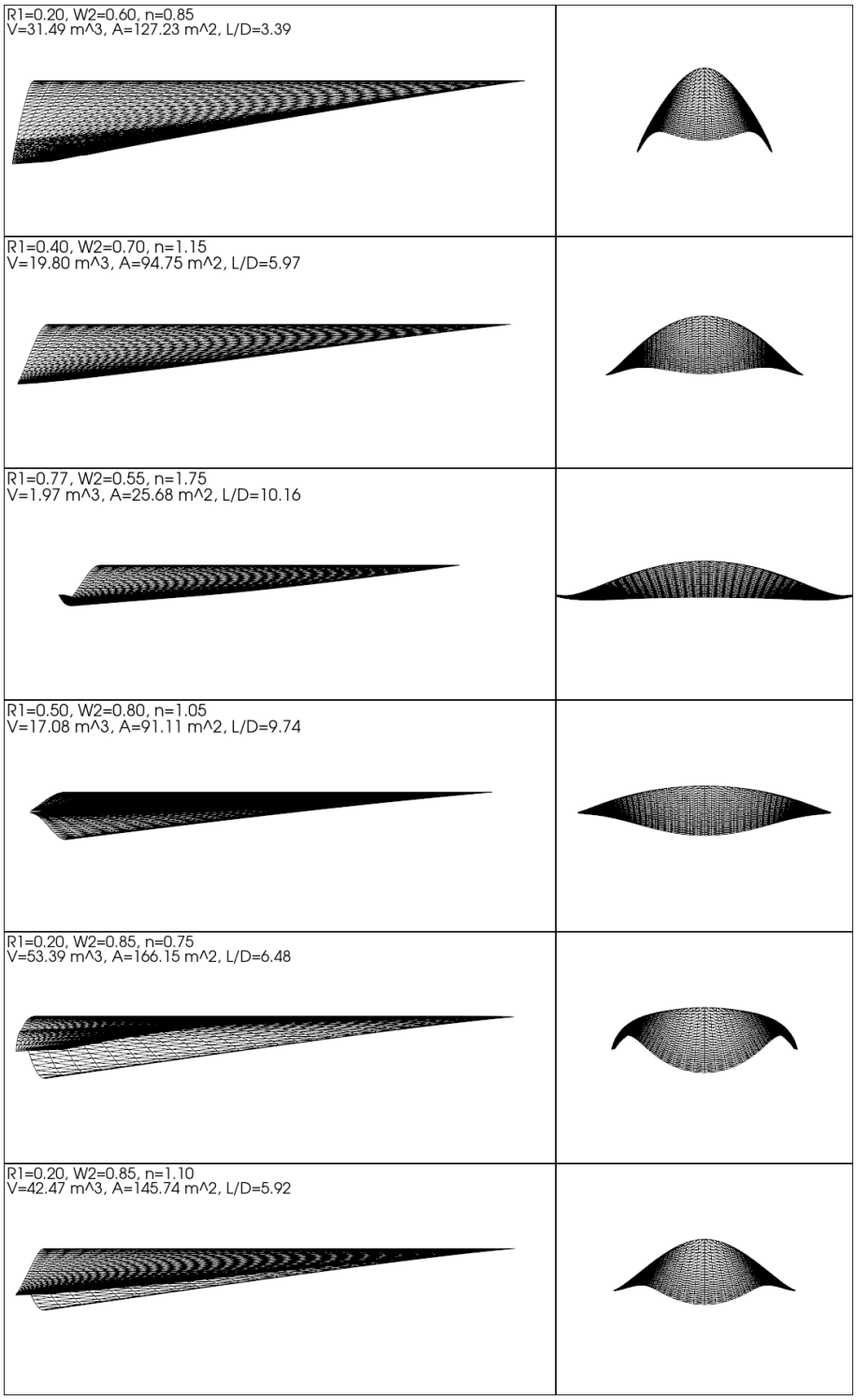


Figure 4: Example waverider geometries generated by the algorithm for different input parameters, using $M_\infty = 6, \beta = 13°, \gamma = 1.4, L = 20$ m. Note that L/D ratios reported above consider inviscid effects only; see the following section for details.

each point on a streamline we look up $V'(\theta)^2 = V_r'^2 + V_\theta'^2$ and compute the dimensional quantities. The stagnation temperature and pressure are constant in the post-shock region; this gives, with freestream T_∞ and p_∞:

$$u = V'(\theta)\sqrt{2\gamma R T_0/(\gamma-1)}, \quad T = T_0\big(1 - V'^2(\theta)\big), \quad M = u/\sqrt{\gamma R T}, \quad p = p_{02}\,(1 - V'^2(\theta))^{\gamma/(\gamma-1)}, \tag{19}$$

where p_{02} is the post-shock stagnation pressure:

$$\frac{p_2}{p_\infty} = \frac{2\gamma M_{n1}^2 - (\gamma-1)}{\gamma+1}, \qquad \frac{p_{02}}{p_\infty} = \frac{p_2}{p_\infty}\left(1 + \tfrac{\gamma-1}{2}M_2^2\right)^{\gamma/(\gamma-1)}. \tag{20}$$

The pressure coefficient follows:

$$C_p^{\text{lower}} = \frac{p/p_\infty - 1}{\frac{\gamma}{2}M_\infty^2} = \frac{2}{\gamma M_\infty^2}\left[\left(\frac{p_{02}}{p_\infty}\right)(1 - V'^2)^{\gamma/(\gamma-1)} - 1\right] \tag{21}$$

To integrate the pressure forces, the lower surface is panelized into triangles with area $\mathrm{d}A$ and outward unit normal $\hat{\mathbf{n}}$. The pressure force on each panel is integrated in coefficient form as

$$\mathrm{d}\mathbf{C}_F = -C_p\,\hat{\mathbf{n}}\,\frac{\mathrm{d}A}{S_{\text{ref}}}, \tag{22}$$

so the total force coefficient vector is $\mathbf{C}_F = \sum \mathrm{d}\mathbf{C}_F$. The reference area is the planform (projection) of the lower surface onto the x-y plane,

$$S_{\text{ref}} = \sum\left|\frac{1}{2}\left[(x_1 - x_0)(y_2 - y_0) - (y_1 - y_0)(x_2 - x_0)\right]\right|, \tag{23}$$

using the projected triangle vertices (x_i, y_i). The coefficient components are reported as $C_L = C_{F,z}$, $C_{D_p} = C_{F,x}$, and $(L/D)_{\text{inviscid}} = C_L/C_{D_p}$.

4.2 Viscous shear contributions via boundary layer integration

To estimate viscous drag on the waverider surface, we use the laminar compressible integral boundary-layer method of Walz [10], exactly following the implementation reported by Bowcutt in his seminal paper on viscous-optimized waverider design [2, 3]. The method assumes that the boundary layer is locally 2D in azimuthal planes, that it is steady, laminar, and attached, that it is in a continuum regime, and furthermore that it develops from a sharp edge. We obtain the boundary layer edge conditions $\{u_e, M_e, \rho_e, T_e\}$ from the external inviscid flow as described above, then march the boundary layer downstream by solving a coupled pair of first-order ordinary differential equations along a streamline on the boundary-layer edge. The model accounts for compressibility, wall-temperature effects, and streamwise pressure-gradient effects through the prescribed edge flow. Flow separation or transition to turbulence are not considered. An isothermal wall at T_w is assumed; radiative equilibrium is not modelled (Bowcutt suggests that this is a reasonable assumption away from the stagnation point).

The two governing equations are the integral momentum equation and the integral mechanical-energy equation, where s is a coordinate along the boundary-layer-edge streamline, and y is the coordinate normal to the surface:

$$\frac{\mathrm{d}Z}{\mathrm{d}s} + \frac{1}{u_e}\frac{\mathrm{d}u_e}{\mathrm{d}s}\,F_1\,Z - F_2 = 0, \qquad \frac{\mathrm{d}W}{\mathrm{d}s} + \frac{1}{u_e}\frac{\mathrm{d}u_e}{\mathrm{d}s}\,F_3\,W - \frac{F_4}{Z} = 0, \tag{24}$$

where Z, W and $\delta_1, \delta_2, \delta_3$ are the thicknesses of the displacement, momentum, and energy boundary

layers, respectively, defined as:

$$Z = \delta_2 \left(\frac{\rho_e u_e \delta_2}{\mu_w} \right) = \frac{\rho_e u_e \delta_2^2}{\mu_w}, \qquad W = \frac{\delta_3}{\delta_2}, \tag{25}$$

$$\delta_1 = \int_0^\delta \left(1 - \frac{\rho u}{\rho_e u_e} \right) \mathrm{d}y, \qquad \delta_2 = \int_0^\delta \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e} \right) \mathrm{d}y, \qquad \delta_3 = \int_0^\delta \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u^2}{u_e^2} \right) \mathrm{d}y, \tag{26}$$

were $(\cdot)_w$ denotes evaluation at the wall. The empirical closures for these equations (copied directly from the textbook [10]) are:

$$F_1 = 3 + 2H - M_e^2 + n \frac{\frac{1}{\mu_w} \frac{\mathrm{d}\mu_w}{\mathrm{d}s}}{\frac{1}{u_e} \frac{\mathrm{d}u_e}{\mathrm{d}s}}, \qquad n = \begin{cases} 0, & T_w = \text{constant}, \\ 1, & T_w \neq \text{constant}, \end{cases} \tag{27}$$

$$F_2 = \frac{2a}{b}, \qquad F_3 = 1 - H + r(\gamma - 1) M_e^2 \left(1 - \frac{\widetilde{\theta}}{W} \right), \qquad F_4 = \frac{2\beta - aW}{b}, \tag{28}$$

and

$$H = \frac{\delta_1}{\delta_2} = b\, H_{12} + r\, \frac{\gamma - 1}{2}\, M_e^2 \left(W - \widetilde{\theta} \right), \tag{29}$$

$$a = 1.7261\, (W^* - 1.515)^{0.7158}, \qquad b = \frac{\delta_2^{(u)}}{\delta_2} = 1 + r\, \frac{\gamma - 1}{2}\, M_e^2 \left(W - \widetilde{\theta} \right) (2 - W), \tag{30}$$

$$r = \sqrt{\mathrm{Pr}}, \qquad \widetilde{\theta} = \frac{T_{aw}(s) - T_w(s)}{T_{aw}(s) - T_e(s)}, \qquad \beta = \beta^{(u)}\, \chi, \tag{31}$$

and

$$H_{12} = 4.0306 - 4.2845\, (W^* - 1.515)^{0.3886}, \qquad T_{aw} = T_e + \frac{r\, u_e^2}{2c_p}, \quad W^* = \frac{\delta_3^{(u)}}{\delta_2^{(u)}} = \frac{W}{\psi}, \tag{32}$$

$$\psi = 1 + \frac{(\psi_{12} - 1) M_e^2}{M_e^2 + \psi_{12}^{-1}/\psi_0}, \qquad \psi_{12} = \frac{2 - \delta_1^{(u)}/\delta}{W^*}\, \widetilde{\theta} + \frac{1 - \delta_1^{(u)}/\delta}{W^* g}\, (1 - \widetilde{\theta}), \tag{33}$$

$$\psi_0 = 0.0144\, (2 - W^*)\, (2 - \widetilde{\theta})^{0.8}, \qquad \frac{\delta_1^{(u)}}{\delta} = 0.420 - (W^* - 1.515)^{0.424\, W^*}, \tag{34}$$

$$g = 0.324 + 0.336\, (W^* - 1.515)^{0.555}, \qquad \beta^{(u)} = 0.1564 + 2.1921\, (W^* - 1.515)^{1.70}, \tag{35}$$

$$\chi = \left\{ 1 + r\, \frac{\gamma - 1}{2}\, M_e^2 \left[1.16\, W^* - 1.072 - \widetilde{\theta}(2W^* - 2.581) \right] \right\}^{0.7} \left\{ 1 + r\, \frac{\gamma - 1}{2}\, M_e^2 (1 - \widetilde{\theta}) \right\}^{-0.7}. \tag{36}$$

Finally, wall viscosity $\mu_w(T_w)$ is given by Sutherland's law,

$$\mu(T) = \mu_0 \left(\frac{T}{T_0^\mu} \right)^{3/2} \frac{T_0^\mu + S}{T + S}, \quad \mu_0 = 1.716 \times 10^{-5}\ \mathrm{Pa\,s},\ T_0^\mu = 273.15\ \mathrm{K},\ S = 110.4\ \mathrm{K}. \tag{37}$$

At the leading edge $s = 0$ the boundary layer has zero thickness, so $Z \to 0$ is singular for the W-equation. We start the integration at a small $s_0 > 0$ from the local self-similar (zero pressure-gradient) state. Setting $\mathrm{d}W/\mathrm{d}s = 0$ in the W-equation with $\mathrm{d}u_e/\mathrm{d}s = 0$ gives the algebraic fixed point $W = 2\beta/a$. With Eq. (32) this is iterated together with the $W \leftrightarrow W^*$ inversion until self-consistent (the incompressible Blasius value $W^* \approx 1.572$ is used as the seed). The matching Z is then $Z(s_0) = (2a/b)\, s_0$, which is the linear Blasius-like growth.

4.2.1 Numerical integration along streamlines

For each streamline, the integration is performed as follows:

1. The streamline is sampled to a uniform grid in arc length s.
2. $\left(u_e, T_e, M_e, \rho_e, \mu_w, \widetilde{\theta}\right)$ are tabulated at each station; linear interpolants are built to evaluate them between stations.
3. The coupled system $\big(Z(s), W(s)\big)$ is integrated with RK45 from the `scipy.integrate.solve_ivp` package.
4. At every RHS evaluation, given the current compressible W and the local edge state, the implicit relation (32) is solved for the self-consistent W^* via `scipy.optimize.brentq`, then a, b, H_{12}, H, $\beta^{(u)}$, χ, β and the four coefficients F_1, F_2, F_3, F_4 are evaluated.

The momentum thickness, wall shear, and skin-friction coefficient are recovered post-integration by:

$$\delta_2 = \sqrt{\frac{Z\,\mu_w}{\rho_e u_e}}, \qquad \tau_w = \frac{a\,\mu_w u_e}{\delta_2}, \qquad c_f = \frac{2\tau_w}{\rho_e u_e^2}. \tag{38}$$

4.2.2 Skin-friction drag integration over the waverider surfaces

On both surfaces, each streamline represents a strip of the wetted surface; strip width $w(s)$ is approximated by the half-distance to each neighbouring streamline at the same fractional arc length (with a single-neighbour doubling at the outermost streamlines). The streamwise friction force on the strip is

$$F_{x,k} \;=\; \int_0^{s_{\mathrm{TE}}} \tau_w(s)\,\big(\mathrm{d}x/\mathrm{d}s\big)\,w(s)\,\mathrm{d}s, \tag{39}$$

evaluated by the trapezoidal rule. The total friction-drag coefficient sums the contributions from both surfaces, each doubled to account for the mirrored half:

$$C_{D_f} \;=\; \frac{2}{q_\infty S_{\mathrm{ref}}}\left(\sum_{k=1}^{N_l} F_{x,k}^{\mathrm{lower}} + \sum_{k=1}^{N_{up}} F_{x,k}^{\mathrm{upper}}\right), \qquad C_{D_f} = C_{D_f}^{\mathrm{lower}} + C_{D_f}^{\mathrm{upper}}, \tag{40}$$

with S_{ref} the planform projection of the lower surface.

4.3 Aerothermal heating and the blunt leading-edge correction

The cone construction produces an aerodynamically sharp leading edge, which is unphysical: stagnation-point convective heating scales as $1/\sqrt{R_n}$ and would be unbounded. A finite blunting radius R_n is required to keep the surface temperature below a chosen material limit T_{allow}. The degree of bluntness must be selected such that maximum expected aerothermal heating is manageable, but should be as small as possible to preserve the aerodynamic performance of the waverider.

The effects of blunting can be significant: local pressure drag is increased, the shockwave may 'leak' over the leading edge to the upper surface, decreasing lift, and a bow shock may form at some standoff distance from the leading edge. Blunting also increases the potential for flow separation and transition to turbulence, which in turn increase drag and heating. That said, for sufficiently performant high-temperature thermal protection system materials, and assuming that a hot aerostructure is employed, the required blunting radius may be very small (at most a few centimeters for the present design space [2, 1]), and thus the vehicle may be considered 'aerodynamically sharp' when considering the macroscopic flowfield. For this reason, we neglect most of the aforementioned effects in the present analysis. Nevertheless, even

a small amount of blunting can induce significant localized drag (up to 8% of total drag [9]), which we do account for via a bluntness correction to the pressure distribution on the leading edge.

Determining the minimum required blunting radius requires an estimate of the aerothermal heating to the vehicle. In our present analysis, we consider a single uniform blunting radius based only on the worst-case heating at the stagnation point. A higher-fidelity analytical treatment of the blunted leading edge may involve allowing for a variable radius and computing heat flux and pressure distributions all along the leading edge, and indeed over the surface of the whole vehicle [9]. However, modern non-equilibrium chemically-reacting CFD analysis of blunted waveriders overwhelmingly supports that heating away from the nose of the vehicle is orders of magnitude lower than the stagnation-point heating [11]; and likewise that the flow is chemically-frozen away from the leading edge [4]. Our simplification is therefore conservative and should not meaningfully underestimate the required blunting radius.

In summary, we treat the blunt leading edge as a circular cylinder of radius R_n swept along the streamline-derived leading edge curve. To determine R_n, we assume that the nose of the waverider is in radiative thermal equilibrium with its environment, and that no active cooling is used; this gives a direct relationship between the blunting radius and the maximum surface temperature. We compute and add the inviscid pressure force contributions of the blunt leading edge to the sharp leading edge result from the previous sections via modified Newtonian impact theory. Since the blunted portions are leading edges, a boundary layer has not yet developed, and we do not account for viscous effects in the bluntness correction.

4.3.1 Minimum blunting radius from aerothermal heating constraints

Convective heating at the stagnation point is estimated from the Sutton-Graves correlation for air in chemical equilibrium [8], which accounts for bow shock heating and exothermic chemical reactions at the stagnation point:

$$\dot{q}_{\text{conv}} = k_{\text{SG}} \sqrt{\frac{\rho_\infty}{R_n}}\, V_\infty^3, \qquad k_{\text{SG}} = 1.83 \times 10^{-4}\ \text{W}\,\text{s}^3\,\text{kg}^{-1/2}\,\text{m}^{-7/2}. \tag{41}$$

This heating is balanced against radiative equilibrium with an emissivity ε, Stefan-Boltzmann constant σ, and background temperature $T_{\text{bg}} \approx 300$ K:

$$\dot{q}_{\text{rad}} = \varepsilon\sigma\left(T_w^4 - T_{\text{bg}}^4\right). \tag{42}$$

Setting $\dot{q}_{\text{conv}} = \dot{q}_{\text{rad}}$ at $T_w = T_{\text{allow}}$ and applying a safety factor $s_f \geq 1$ gives

$$R_{n,\text{min}} = \rho_\infty \left(\frac{s_f\, k_{\text{SG}}\, V_\infty^3}{\varepsilon\sigma\left(T_{\text{allow}}^4 - T_{\text{bg}}^4\right)} \right)^2. \tag{43}$$

The maximum allowable temperature, T_{allow} and the emissivity ε are material properties of the thermal protection system, and the safety factor s_f accounts for uncertainties in the heating estimate and material properties. For a given material choice, Eq. (43) gives a direct relationship between the blunting radius and the maximum expected aerothermal heating.

4.3.2 Modified-Newtonian force along a swept leading edge

To determine the contribution of the blunted leading edge to the overall aerodynamic forces, we apply modified-Newtonian theory. At each point along the leading edge, the velocity $\mathbf{u}_\perp$ and Mach number,

$M_\perp$, perpendicular to the leading edge (normal to the local tangent) are computed as:

$$\hat{\mathbf{u}}_\perp = \frac{\hat{\mathbf{u}}_\infty - \sin\Lambda\,\hat{\mathbf{t}}}{\cos\Lambda}, \qquad M_\perp = M_\infty \cos\Lambda, \tag{44}$$

where $\Lambda = \hat{\mathbf{t}}\cdot\hat{\mathbf{u}}_\infty$ is the local sweep angle between the leading edge tangent and the freestream direction, and $\hat{\mathbf{t}}$ is the local unit tangent vector. In the plane perpendicular to $\hat{\mathbf{t}}$, the cylinder cross-section is a circle of radius R_n, parameterised by an angle ϕ measured from the stagnation line. The modified-Newtonian distribution is

$$C_p(\phi) = C_{p,\max}(M_\perp)\,\cos^2\Lambda\,\cos^2\phi, \qquad |\phi| \le \pi/2 \text{ (windward)}, \tag{45}$$

and zero on the leeward side. $C_{p,\max}$ follows from the normal-shock stagnation-pressure construction:

$$C_{p,\max}(M_\perp) = \frac{p_{02}/p_1 - 1}{\frac{1}{2}\gamma M_\perp^2}, \tag{46}$$

where p_{02}/p_1 is built from the standard normal-shock and isentropic relations applied to $M_\perp$. Integrating the pressure load $-C_p\,q_\infty\,R_n\,\hat{\mathbf{n}}\,\mathrm{d}\phi\,\mathrm{d}s$ over the windward half exploits $\int_{-\pi/2}^{\pi/2}\cos^3\phi\,\mathrm{d}\phi = 4/3$ and gives a closed-form resultant force per unit leading edge arc length,

$$\frac{\mathrm{d}\mathbf{F}}{\mathrm{d}s} = \tfrac{4}{3}\,q_\infty\,R_n\,C_{p,\max}(M_\perp)\,\cos^2\Lambda\,\hat{\mathbf{u}}_\perp. \tag{47}$$

The transverse component (in the $\hat{\mathbf{t}}\times\hat{\mathbf{u}}_\perp$ direction) vanishes by symmetry of $\cos^2\phi$ about $\phi = 0$.

The leading-edge curve $\{(X_k, Y_k, Z_k)\}_{k=1}^N$ from Section 3.2 is differenced to obtain segment tangents $\hat{\mathbf{t}}_k$ and arc lengths Δs_k. Eq. (47) is evaluated at every segment and summed,

$$\mathbf{F}_{\mathrm{LE}} = \sum_k \tfrac{4}{3}\,q_\infty\,R_n\,C_{p,\max}(M_{\perp,k})\,\cos^2\Lambda_k\,\hat{\mathbf{u}}_{\perp,k}\,\Delta s_k. \tag{48}$$

Because the discretised leading edge spans the full vehicle ($y \in [-y_{\mathrm{up}}, +y_{\mathrm{up}}]$) no symmetry doubling is required. The resultant lift- and drag-coefficient increments are

$$\Delta C_L = \frac{F_{\mathrm{LE},z}}{q_\infty S_{\mathrm{ref}}}, \qquad \Delta C_D = \frac{F_{\mathrm{LE},x}}{q_\infty S_{\mathrm{ref}}}. \tag{49}$$

Finally, the total drag coefficient is $C_D = C_{D_p} + C_{D_f} + C_{D_B}$, and the total lift-to-drag ratio is $L/D = C_L/C_D$.

4.4 Viscous Interaction and Shock Wave-Boundary Layer Interaction (SWBLI)

In the hypersonic regime, the immense aerodynamic heating and flow deceleration near the vehicle surface can cause the boundary layer to grow exceptionally thick, leading to viscous pressure interaction which can significantly increase skin friction and heating rates. For our waverider, which relies on a perfectly tailored geometry to keep its bow shock attached exactly to its leading edge, a thick boundary layer can also push the shock wave outward, causing undesirable 'leakage' of high-pressure air from the lower surface, further degrading lift [1]. A similarly important effect is shock wave-boundary layer interaction (SWBLI), which occurs when a shock wave impinges on the boundary layer, causing it to thicken and potentially separate.

In the present work, we treat neither effect. Detailed exploration of the waverider design space by [4] shows that for a 60-meter-long hypersonic waverider, viscous pressure interactions have significant influence (greater than 5%) on L/D only for certain Mach number-altitude combinations, highlighted in light blue on Fig. 5. The region of interest for the present work (Mach 4 to 6, altitudes 50,000 to 70,000 ft) is far outside the region of significant interaction, which justifies our neglect of viscous pressure interactions.

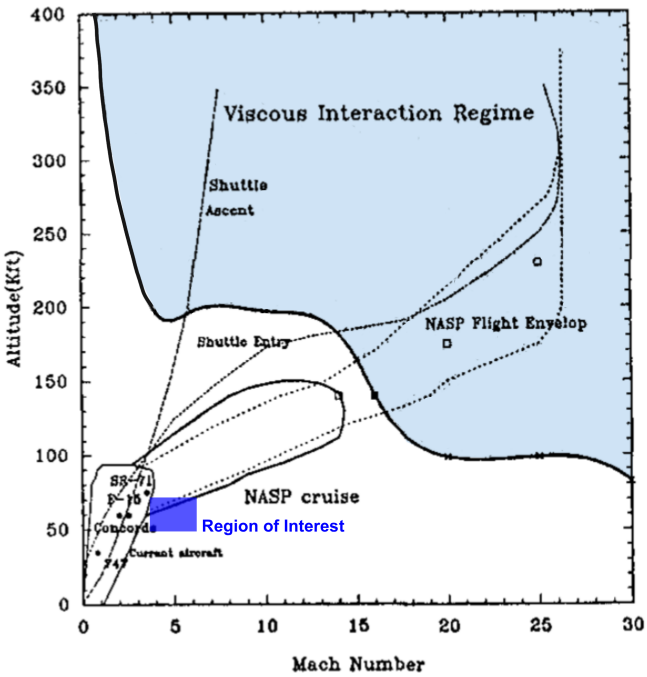


Figure 5: Notional velocity-altitude plot showing the region (light blue) where shock wave-boundary layer interactions significantly influence L/D ($> 5\%$ change) for a 60-m hypersonic waverider. The region of interest for the present work (Mach 4 to 6, altitudes 50,000 to 70,000 ft) is indicated in dark blue; it is far outside the region of significant interaction. Adapted from [4, 1].

As for the shock-wave-boundary layer interaction: by design, all shocks produced by our waverider are directed outward and away from the vehicle, and so SWBLI is avoided. We are afforded the luxury of such a design because the practical engineering and manufacturing considerations that may lead to the necessary protrusions that would produce such impinging shocks (*e.g.,* engine nacelles, cockpit windows, stabilizing fins, ...) are outside the scope of the present investigation. Nevertheless, we acknowledge that these effects can be significant, and that they should be considered in a more comprehensive analysis.

5 Material selection and thermal protection system design

6 Propulsion system design

7 Selection of Optimized Geometry

In Sections 3 and 4, we established an algorithm for the geometric construction of the waverider and the methods for computing its aerothermodynamic performance. In this section, we seek to optimize the design parameters to maximize the aerodynamic efficiency (the total lift-to-drag ratio L/D) at the cruise condition, subject to coupled geometric design constraints described in Sections 1, 2 and 6. To achieve this, we perform a global optimization of all geometric parameters $(L, \beta, R_{1,\text{frac}}, W_{\text{frac}}, n)$ to maximize L/D at the cruise condition.

7.1 Optimization Method

The formal problem is posed as:

$$\begin{aligned} \min_{\mathbf{x}} \quad & J(\mathbf{x}) = -\frac{L}{D}(\mathbf{x}) \\ \text{subject to} \quad & g_1(\mathbf{x}) = H_{\text{req}} - H(\mathbf{x}) \leq 0 \\ & g_2(\mathbf{x}) = A_{\text{req}} - A(\mathbf{x}) \leq 0 \\ & g_3(\mathbf{x}) = V_{\text{cabin}} + V_{\text{fuel}}(\mathbf{x}) - V(\mathbf{x}) \leq 0 \\ \text{with bounds} \quad & \mathbf{x}_{\min} \leq \mathbf{x} \leq \mathbf{x}_{\max} \end{aligned} \tag{50}$$

where $\mathbf{x} = [\beta, L, R_{1,\text{frac}}, W_{\text{frac}}, n]^\top$ are the design variables, $J(\mathbf{x})$ is the objective function to minimize (negative L/D), and $g_i(\mathbf{x})$ are the constraint functions that must be satisfied. The bounds on the design variables are defined by $\mathbf{x}_{\min}$ and $\mathbf{x}_{\max}$: the shock-angle bounds are set from the Mach angle, $\beta_{\min} = \max\left(\sin^{-1}(1/M_\infty) + 0.5°,\, 1°\right)$ and $\beta_{\max} = 18°$, while the trailing-edge bounds are taken from their admissible ranges (see Section 3.2).

To solve the optimization proble, we use a differential evolution algorithm [7] from the SciPy library, `scipy.optimize.differential_evolution`, which is a powerful global optimization method that does not require gradient information and is well-suited for non-convex problems with multiple local minima. The algorithm iteratively evolves a population of candidate solutions by applying mutation and crossover operations, guided by the objective function evaluations, to explore the design space effectively. To enforce the geometric constraints, we use a predictor-corrector approach within the optimization loop, as a unique generating cone length L satisfies all constraints for a given choice of trailing-edge parameters. To determine L, we first generate a unit-length waverider ($L = 1$) with the same M_∞, γ, β, and trailing-edge parameters and compute its cabin volume $V_{L=1}$, area $A_{L=1}$, and height $h_{L=1}$ (the predictor step). Since the geometry scales linearly with L, the required length is

$$L = \max\left(L_{\text{vol}},\, L_{\text{area}},\, L_{\text{height}}\right), \tag{51}$$

where

$$L_{\text{vol}} = \left(\frac{V_{\text{cabin}} + V_{\text{fuel}}}{V_{L=1}}\right)^{1/3}, \qquad L_{\text{area}} = \left(\frac{A_{\text{req}}}{A_{L=1}}\right)^{1/2}, \qquad L_{\text{height}} = \frac{h_{\text{req}}}{h_{L=1}}. \tag{52}$$

We then update the waverider geometry with the computed L and evaluate the objective function (the corrector step). This approach ensures that all geometric constraints are satisfied for each candidate solution evaluated during the optimization process.

7.2 Optimized waverider results

The results of the optimization routine described above lead to the converged geometry shown in Fig. 6 that maximizes total L/D at the cruise condition while satisfying all geometric constraints. As detailed in Table 1, the optimizer selected a wide concave configuration with a total length of 54.77 m and a generating shock angle of $\beta = 11.4°$. The specific combination of trailing edge parameters ($R_{1,\text{frac}} = 0.835$, $W_{2,\text{frac}} = 0.360$, $n_{\text{shape}} = 1.040$) yields a profile where the active constraint is cabin height; generated volume and area are both in excess of the requirements. The resulting $L/D = 6.58$ is staggeringly large for hypersonic flight. The geometry and aerodynamic performance metrics of the opimized design are summarized in Table 1.

Figs. 7 and 8 display the Mach number and temperature contours, respectively, both on the vehicle and on a series of planes intersecting the flow, obtained by post-processing the inviscid Taylor-Maccoll solutions. They illustrate how the waverider 'surfs' the conical shockwave with a bow shock that remains perfectly attached to the leading edge at the Mach 6 cruise condition. This attached shock ensures that the high

Table 1: Optimized waverider design parameters and aerodynamic performance

Category	Parameter	Value
Shock Conditions	Post-shock Mach (M_2)	5.625
	Deflection Angle (θ)	2.692°
	Cone Angle	6.231°
	Virtual Cone Length	331.911 m
Trailing Edge	$R_{1,\text{frac}}$	0.835
	$W_{2,\text{frac}}$	0.360
	n_{shape}	1.040
	Shock Angle (β)	11.397°
Geometry	Vehicle Length	54.765 m
	Maximum Height (H)	3.000 m
	Wetted Area (S_{wet})	3483.782 m^2
	Total Volume (V)	1874.196 m^3
	Mesh Resolution	6728 Triangles
Aerodynamic Coefficients	Mean C_p	0.0210
	Lift Coefficient (C_L)	0.0214
	Inviscid Drag (C_D)	0.0011
	Lower Skin Friction (C_{D_f})	0.0006
	Upper Skin Friction (C_{D_f})	0.0006
Blunt Leading-Edge Correction	Thermal Limit (T_{allow})	2500.0 K
	Minimum Radius (R_n)	51.42 mm
	Bluntness Lift Penalty (ΔC_L)	0.0003
	Bluntness Drag Penalty (ΔC_D)	0.0010
Totals	**Total Lift (C_L)**	**0.0217**
	Total Drag (C_D)	**0.0033**
	Inviscid L/D	19.162
	Viscous L/D (Sharp)	9.320
	Total L/D	**6.579**

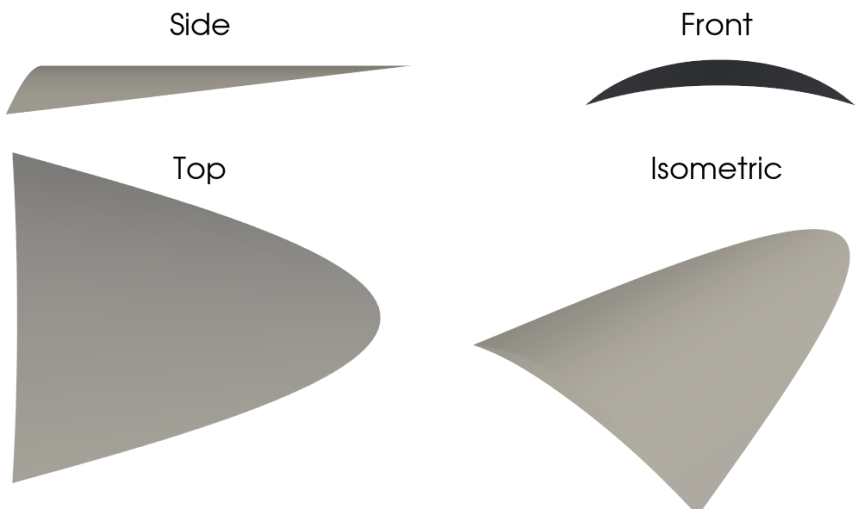


Figure 6: Geometry views of the optimized waverider designs at cruise conditions.

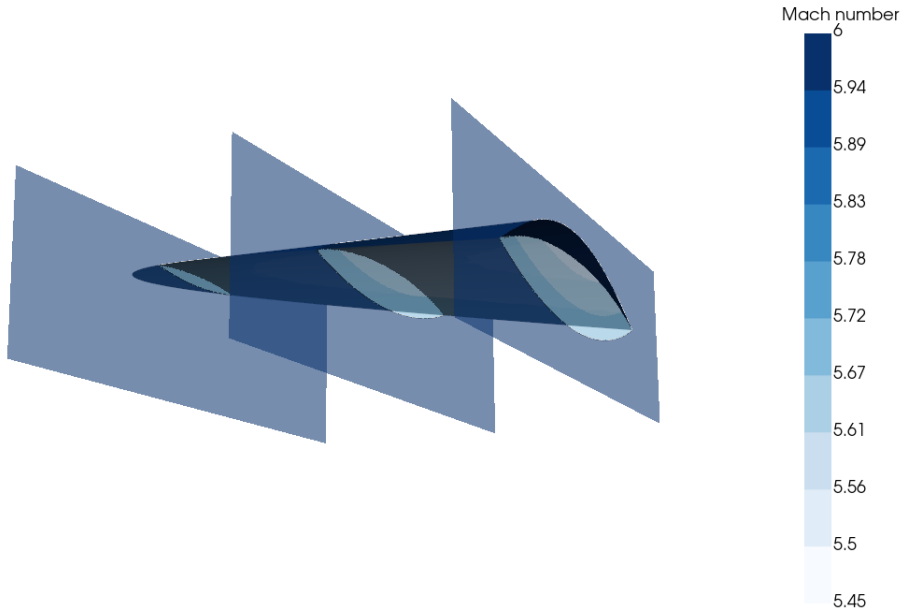


Figure 7: Mach number contours on the vehicle surface and on three planes normal to the free stream for the optimized waverider designs at cruise conditions. The flowfield solutions show attached shockwave.

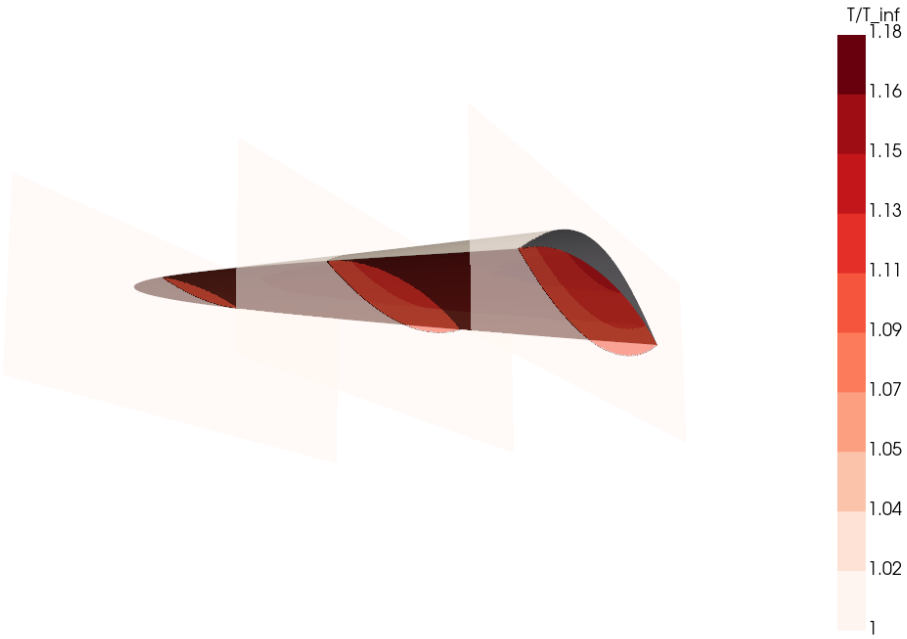


Figure 8: Temperature contours on the vehicle surface and on three planes normal to the free stream for the optimized waverider designs at cruise conditions. The flowfield solutions show attached shockwave.

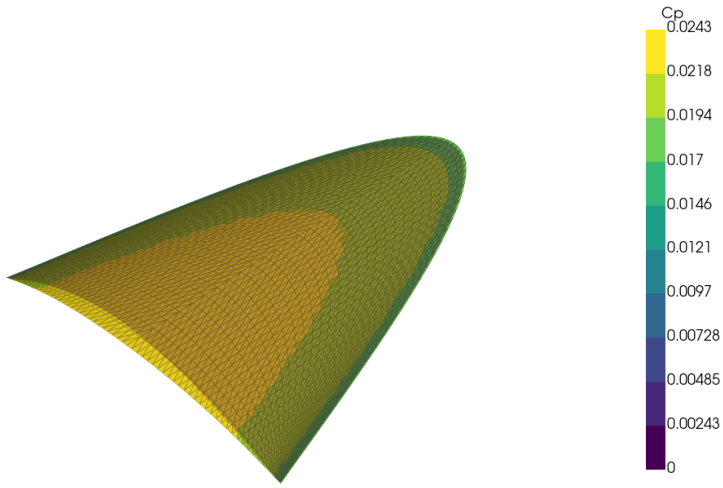
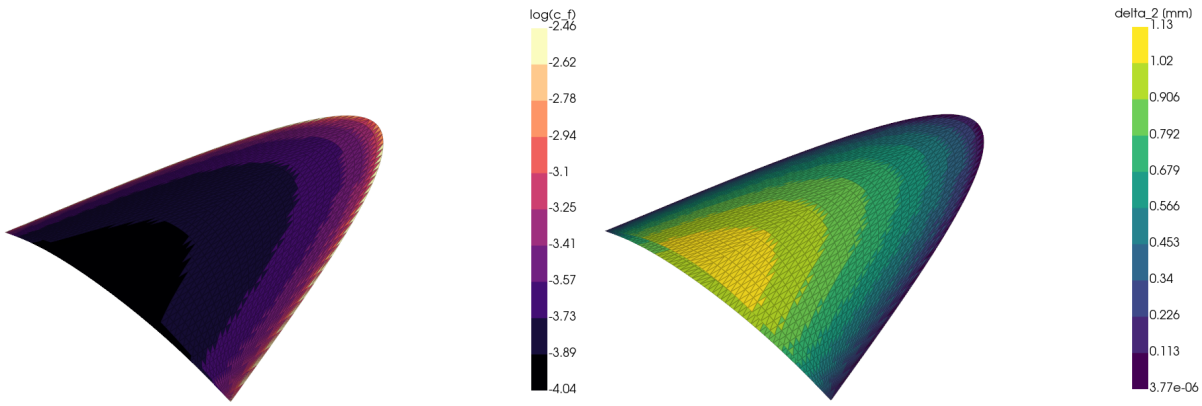


Figure 9: Pressure coefficient contours on the surface of the optimized waverider designs at cruise conditions.



(a) Skin friction coefficient **(b)** Momentum boundary layer thickness

Figure 10: Solution of the integrated Walz boundary layer correlations on the surface of the vehicle. The boundary layer is laminar and attached, with maximum skin friction at the leading edge.

aerodynamic pressure on the lower surface, shown in Fig. 9, does not spill over to the upper surface. Based on these purely inviscid considerations, the waverider achieves a lift coefficient of $C_L = 0.0214$ and a pressure drag coefficient of $C_D = 0.0011$, yielding an inviscid lift-to-drag ratio of $(L/D)_{\text{inviscid}} \approx 19.16$.

Fig. 10 displays the momentum boundary layer thickness and local skin friction coefficient which result from solving the Walz boundary layer integral equations. Integrating the coefficient shows that the boundary layer generates substantial drag across the vehicle's large wetted area ($S_{\text{wet}} \approx 3484$ m^2), contributing an additional $\Delta C_D = 0.0012$ in drag. Furthermore, to survive the intense localized aerodynamic heating at the leading edge without exceeding the refractory composite limit of $T_{\text{allow}} = 2500$ K, a minimum blunting radius of $R_n = 51.42$ mm was required. This geometric blunting penalty adds a further $\Delta C_D = 0.0010$ in wave drag. Ultimately, the accumulation of skin friction and blunt-body drag reduces the physically realistic efficiency of the waverider to a total $L/D = 6.58$.

8 Limitations

"It is really not foreseeable that an 'optimized' calculated shape could do anything more than give a guide to the designer. [… The designer] would really be a little embarrassed to be offered a perfect aerodynamic shape, which he would then have to carve holes in, add fairings, and so on, in order to satisfy such mundane requirements as that the pilot should be able to see where he is going or that people have somewhere convenient to get in and out." — Phillip Roe, 1970

9 References

References

[1] J.D. Anderson. *Hypersonic and High-temperature Gas Dynamics*. AIAA education series. American Institute of Aeronautics and Astronautics, Incorporated, 2019. ISBN: 9781624105142.

[2] Kevin G. Bowcutt. "Optimization of hypersonic waveriders derived from cone flows - including viscous effects". PhD thesis. University of Maryland, College Park, 1986, p. 256. ISBN: 979-8-206-35031-9.

[3] Kevin G. Bowcutt, John D. Anderson, and Diego Capriotti. "Viscous optimized hypersonic waveriders". In: *25th AIAA Aerospace Sciences Meeting*. Reno, NV: American Institute of Aeronautics and Astronautics, 1987, pp. 1–18. DOI: 10.2514/6.1987-272.

[4] Jr. John Anderson, Jinhwa Chang, and Thomas Mclaughlin. "Hypersonic waveriders — Effects of chemically reacting flow and viscous interaction". In: *30th Aerospace Sciences Meeting and Exhibit*. American Institute of Aeronautics and Astronautics, pp. 1–13. DOI: 10.2514/6.1992-302.

[5] J. G. Jones et al. "A method for designing lifting configurations for high supersonic speeds, using axisymmetric flow fields". In: *Ingenieur-Archiv* 37.1 (Jan. 1968), pp. 56–72. ISSN: 1432-0681. DOI: 10.1007/BF00532683.

[6] J. W. Maccoll. "The conical shock wave formed by a cone moving at a high speed". In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 159.898 (Apr. 1937), pp. 459–472. ISSN: 0080-4630. DOI: 10.1098/rspa.1937.0083.

[7] Rainer Storn and Kenneth Price. "Differential Evolution – A Simple and Efficient Heuristic for global Optimization over Continuous Spaces". In: *Journal of Global Optimization* 11.4 (Dec. 1997), pp. 341–359. ISSN: 1573-2916. DOI: 10.1023/A:1008202821328.

[8] Kenneth Sutton and Roger Graves. *A general stagnation-point convective heating equation for arbitrary gas mixtures*. Tech. rep. Hampton, VA: NASA Langley Research Center, 1971. URL: https://ntrs.nasa.gov/citations/19720003329.

[9] D. Vanmol. "Heat transfer characteristics of hypersonic waveriders with an emphasis on the leading edge effects". University of Maryland, College Park, 1992, p. 123. URL: https://ntrs.nasa.gov/citations/19920012972.

[10] A. Walz. *Boundary Layers of Flow and Temperature*. Cambridge, MA: MIT Press, 1969.

[11] Erik Zhou. *Waverider-Generator*. https://github.com/ExusiaiVAL/Waverider-Generator. 2025.